

AI61002 : Deep Learning Foundations and Applications

Spring 2026

Assignment 3

(20 Marks)

Instructions: Complete the tasks listed below, ensuring you follow the specified instructions for each. Provide step-by-step outputs in the Jupyter Notebook (.ipynb file). All relevant code snippets, visualizations, and explanations must be included within the notebook. Submit the required files as per the submission guidelines provided.

Question 1 - 3 marks

Load the Amazon Product Reviews dataset from Kaggle and preprocess the review text using NLTK by removing stopwords, removing punctuation, performing tokenization, and converting tokens into numerical form using torchtext vocabulary.

Question 2 - 2 marks

Show the distribution of the review ratings (1,2,3,4,5) and split the dataset into training, validation, and test sets (70%, 15%, 15%) with padding applied for equal sequence length.

Question 3 - 5 marks

Given that the task is to predict the integer ratings 1–5 from the reviews, determine whether the problem should be modeled as a classification or regression problem, and design a GRU-based model using PyTorch with nn.GRUCell (instead of nn.GRU) to predict the rating from the product reviews by choosing the number of GRU layers and hidden state dimension for each layer to achieve good prediction performance.

Question 4 - 2 marks

Show the summary of the model architecture designed in the previous question, print the number of trainable parameters of the model, and design a suitable loss function (classification or regression) depending on how you modeled the problem.

Question 5 - 2 marks

Train the model using the Adam optimizer and show the variation of training and validation losses on the training progress.

Question 6 - 2 marks

Perform prediction of the ratings on the test set using the trained model. Report the correlation between the ground truth ratings and the predicted ratings on the test set using Spearman's Rank Order Correlation Coefficient (SROCC).

Question 7 - 4 marks

Re-design the same model using nn.GRU, train and evaluate it on the test set. Report the SROCC of the predicted ratings on the test set with the ground truth ratings, and compare it to that of the model designed using nn.GRUCell.

Submission Guidelines

- The content that you submit must be your individual work.
- Submit both .py and .ipynb file formats. Both submissions are mandatory for credit.
- Name the files as "<roll_number>_assignment_1.py" and "<roll_number>_assignment_1.ipynb". Submit them in a single zip file named "<roll_number>_assignment_1.zip". For example, if your roll number is 23AI91R01, the submission file should be named 23AI91R01_assignment_1.zip (Note: do not submit .rar files).
- Ensure all code is well-commented and explanations are included using text cells in the Jupyter Notebook.
- Submit your assignment through Moodle before the deadline. Late submissions or email submissions will not be accepted.

TA for this assignment: Chhavi Chaudhary
For any queries, email : chhavi073@kgpian.iitkgp.ac.in